

Technology #52: Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Technology Domain: fluidics_hydraulic

Mechanism: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Apply a micro■textured or hydrophilic anti■adhesive coating to the intraocular lens surface that resists capsule adhesion under mechanical stress, thereby preventing fluid retention and subsequent microbial colonization.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

Next Experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

Buyer Receives: canonical invention specification (with epistemic classes); engineering specification (this document, all blocks classed); design input/output graph with explicit IDs; engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification); killer-experiment definition and its EIG basis; evidence custody references and hashes; unresolved-uncertainty register; domain equation set (symbolic; sources and applicability judgments identified)

Buyer Must Build: actual geometry and detailed design; sourced engineering parameters and acceptance thresholds; physical prototypes; all verification and validation results; manufacturing process qualification; regulatory pathway determination and submissions

Transfer Posture: Sponsored Validation

Target Buyer: domain-appropriate technology buyer (to be identified by portfolio owner)

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